

10 - Updating record in the database

The UPDATE SQL statement is used to modify the data in one or more rows of a table.

The `executeUpdate()` method is used for UPDATE operations, and it returns an integer indicating the number of rows affected.

Steps For Updating the Data:

1. Create an SQL UPDATE Statement:

```
String sql = "UPDATE studentInfo SET sage = 24 WHERE id = 2";
```

2. Execute the SQL Statement:

Use the Statement object's `executeUpdate()` method to execute the UPDATE statement.

```
int rowAffected = statement.executeUpdate(sql);
```

3. Check If the Update Was Successful:

After executing the UPDATE statement, check the `rowAffected` variable to determine if the update was successful.

```
if (rowAffected > 0) {  
    System.out.println("Update successful! Number of rows affected: " +  
        rowAffected);  
} else {  
    System.out.println("Update failed. No rows were affected.");  
}
```

□ While values are hard-coded in the SQL statement in the given example, it is also possible to pass them dynamically at runtime using the `PreparedStatement`, which will be discussed later.

Code:

```
import java.sql.*;

public class LaunchApp2 {

    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/yourDatabaseName";

        String username = "yourUsername";

        String password = "yourPassword";

        String sql = "UPDATE studentInfo SET sage = 24 WHERE id = 2";

        Connection conn = null;

        Statement stmt = null;

        try {

            conn = DriverManager.getConnection(url, username, password);

            stmt = conn.createStatement();

            int rowAffected = stmt.executeUpdate(sql);

            if (rowAffected > 0) {

                System.out.println("Update successful! Number of rows affected: " + rowAffected);

            } else {

                System.out.println("Update failed. No rows were affected.");

            }

        } catch (SQLException e) {

            e.printStackTrace();

        } finally {
```

```
try {  
    if (stmt != null) stmt.close();  
    if (conn != null) conn.close();  
} catch (SQLException e) {  
    e.printStackTrace();  
}  
}  
}
```